Total No. of Questions : 5]

SEAT No. :

PA-1031

[Total No. of Pages : 3

[5902]-61

T.Y. B.Sc. (Semester - VI)

COMPUTER SCIENCE

CS-361 : Operating System - II

(2019 Pattern) (CBCS)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) *All questions are compulsory.*
2) *Figures to the right indicate full marks.*

Q1) **Attempt any eight of the following :** **[8 × 1 = 8]**

a) What is request edge?

b) What is safe state?

c) Write the names of any two disk allocation methods of disk space.

d) List disk performance parameters.

e) Define distributed system.

f) What is size scalability?

g) List the different architectural styles of distributed operating systems.

h) What is kernel?

i) What is RISC in ARM architecture?

j) Write any two special service requirements of mobile operating system.

P.T.O.

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Q2) **Attempt any four of the following :** **[4 × 2 = 8]**

a) Write the difference between SCAN & LOOk disk scheduling algorithms.

b) Define seek time & rotational latency.

c) Explain features of mobile operating system.

d) Give a comparative study of any four points of Android mobile operating system and Apple i05 mobile operating system.

e) Write a short note on centralized organization of system architecture.

Q3) **Attempt any two of the following :** **[2 × 4 = 8]**

a) Explain any two deadlock prevention strategies.

b) Explain sequential access & Direct access methods for a file.

c) Write a short note on cloud computing system.

Q4) **Attempt any two of the following :** **[2 × 4 = 8]**

a) Consider following snapshot of the system. A, B, C, D are the resource types. Answer the following questions using Banker's algorithm.

i) What are the contents of Need matrix/array?

ii) If the system is in the safe state, give the safe sequence.

	Allocation				Max				Total			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2	1	5	2	0
P_1	1	0	0	0	1	7	5	0				
P_2	1	3	5	4	2	3	5	6				
P_3	0	6	3	2	0	6	5	2				
P_4	0	0	1	4	0	6	5	6				

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b) Explain any four file operations.

c) Explain the design goals of distributed systems.

Q5) **Attempt any one of the following :** **[1 × 3 = 3]**

a) What is total head movement for First-Come First-Served (FCFS) scheduling for the disk queue with requests for I/O to blocks on cylinders 98, 183, 37, 122, 14, 124, 65, 67 in that order, If the disk head is initially at cylinder 53.

b) Explain the special constraints & requirements of mobile operating system.

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Total No. of Questions : 5]

SEAT No. :

P-6398

[Total No. of Pages : 2

[6155]-61
T.Y. B.Sc. (Computer Science)
CS361 : OPERATING SYSTEMS - II
(2019 Pattern) (Semester-VI)

Time : 2 Hour] *[Max. Marks : 35*

Instructions to the candidates :

1) ***All questions are compulsory.***
2) ***Figures to the right indicates full marks.***

Q1) Attempt any Eight of the following. **[8×1=8]**

a) List all dead lock recovery methods.

b) List file system free space management techniques.

c) What is disk scheduling?

d) What is deadlock?

e) Define object-based architecture.

f) List system architectures.

g) List any Four commercial mobile operating systems.

h) What is kernel?

i) What are the features of mobile operating systems?

j) List the types of distributed systems.

Q2) Attempt any Four of the following. (Out of Five) **[4×2=8]**

a) What are the goals of distributed systems.

b) Differentiate scan and c-scan disk scheduling.

c) What is ARM?

d) What is native code?

e) Explain resource - allocation graph with example.

P.T.O.

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Q3) Attempt any two of the following. (Out of Three) **[2×4=8]**

a) Consider the following snapshot of system A,B,C,D are the resource types.

	Allocation			
	A	B	C	D
P_0	0	0	1	2
P_1	1	0	0	0
P_2	1	3	5	4
P_3	0	6	3	2
P_4	0	0	1	4

	MAX			
	A	B	C	D
P_0	0	0	1	2
P_1	1	7	5	0
P_2	2	3	5	6
P_3	0	6	5	2
P_4	0	6	5	6

Available			
A	B	C	D
1	5	2	0

Answer the following questions using Banker's Algorithm:

i) What are the contents of need array?

ii) Is the system in safe state? If yes give safe sequence.

iii) If a request from P_1 arrives for (0,4,2,0) can it be granted immediately?

b) Explain the architecture of Android OS.

c) Explain access methods of file system management.

Q4) Attempt any two of the following. (Out of Three) **[2×4=8]**

a) Differentiate Desktop OS and Mobile OS.

b) Explain the necessary conditions of deadlock with suitable example and diagram.

c) Write a short note on directory structure.

Q5) Attempt any ONE of the following. (Out of two) **[1×3=3]**

a) Consider following work queue : 23, 89, 132, 42, 187 & show schedule using following algorithms :

i) SSTF

ii) SCAN

iii) C-LOOK

Also find total head movements in each algorithm.

b) Differentiate between Android OS and iphone OS.

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Total No. of Questions : 5]

PB1891

SEAT No. :

[Total No. of Pages : 2

[6237]-601
T.Y.B.Sc.
COMPUTER SCIENCE
CS-361 : Operating Systems - II
(Revised 2019 Pattern) (Semester-VI)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***Figures to the right indicate full marks.***

Q1) Attempt any Eight of the following questions. **[8×1=8]**

a) What is safe sequence?

b) What is grid computing?

c) Write file access methods.

d) Define rotational Latency.

e) What are the different types of distributed system?

f) List any four file attributes.

g) List any two important features of an Android mobile OS.

h) "Hybrid cloud is a combination of the public cloud and the private cloud". Comment (True/False)

i) State all the necessary conditions for a deadlock to occur.

j) List any four special constraints of mobile operating system.

Q2) Attempt any four of the following questions. **[4×2=8]**

a) What is C-Scan and C-look? Compare them.

b) What is mobile operating system? What are it's responsibilities?

c) Explain deadlock prevention strategies.

d) Explain different methods for handling free-space list in file system.

e) Explain cloud computing.

P.T.O.

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Q3) Attempt any Two of the following questions. **[2×4=8]**

a) Compare Desktop OS and Mobile OS.

b) What are the operations performed on files?

c) Consider given snapshot of system. A system has 5 processes and 3 types of resources A,B,C

Allocation			
	A	B	C
P0	2	3	2
P1	4	0	0
P2	5	0	4
P3	4	3	3
P4	2	2	4

Max		
A	B	C
9	7	5
5	2	2
11	0	4
4	4	4
6	5	5

Available		
A	B	C
3	3	2

Answer the following questions using Banker's Algorithm.

i) What are the contents of need matrix?

ii) Is the system in a safe state? If yes find safe sequence.

Q4) Attempt any two of the following. **[2×4=8]**

a) Define P_2P architecture of distributed OS.

b) Consider the following sets P, R and E

$P = [P_1, P_2, P_3]$

$P = [R_1, R_2, R_3, R_4]$

$E = [P_1 \rightarrow R_1, P_2 \rightarrow R_3, R_1 \rightarrow P_2, R_2 \rightarrow P_2, R_2 \rightarrow P_1]$

Also consider the following number of instances per resource type.

i) One instance of resource type R_1 and R_2.

ii) Two instance of resource type R_2.

iii) Three instance of resource type R_4.

Construct the resource allocation graph for the above problem. Check whether the system is in deadlock.

c) What is directory? What are it's type? Explain two of them.

Q5) Attempt any one of the following. **[1×3=3]**

a) What is total head movement for FcFs and SSTF scheduling for the disk queue with requests for I/o to blocks on cylinders. 176, 79, 34, 60, 92, 11, 41, 114 in that order. If the disk head is initially at cylinder 50.

b) What is client server system. Also state it's advantages.

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Total No. of Questions : 5]

SEAT No. :

PC-1587

[Total No. of Pages : 3

[6328]-61
T.Y. B.Sc.
COMPUTER SCIENCE
CS-361 : Operating Systems-II
(Rev. 2019 Pattern) (Semester - VI)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) ***All questions are compuslory.***
2) ***Figures to the right indicates full marks.***

Q1) Attempt any Eight of the following : **[8 × 1 = 8]**

a) List any four file attributes.

b) Define starvation.

c) What do you mean by full stroke lime ?

d) Define Distributed systems.

e) List the limitations of single level directory.

f) What is mobile operating system ?

g) Kernel is core of any operating system. Justify True/False.

h) Define deadlock.

i) What do you mean by disk Bandwidth ?

j) What do you mean by Grid Computing ?

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P.T.O.

Q2) Attempt any Four of the following : **[4 × 2 = 8]**

a) Write features of mobile operating system.

b) List the different types of Distributed systems.

c) Explain any two Disk performance parameters.

d) Explain resource allocation Graph.

e) What do you mean by absolute path and relative path.

Q3) Attempt any Two of the following : **[2 × 4 = 8]**

a) Explain Design goals of distributed systems.

b) Write short note on Disk Mangement.

c) Consider the following snapshot of the system A, B, C, D are the resource types.

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2	1	4	2	0
P_1	1	1	0	0	1	7	5	0				
P_2	1	3	5	4	2	3	5	6				
P_3	0	6	3	2	0	6	5	2				
P_4	1	0	1	4	1	6	5	6				

Answer the following questions using Banker's Algorithm.

i) What are the contents of need array.

ii) If the system is in safe state give the safe sequence.

iii) If the request from process P_1 arrived for (0, 4, 2, 0) can it be granted immediately.

Q4) Attempt any Two of the following : **[2 × 4 = 8]**

a) Explain the necessary conditions for a Deadlock to occur.

b) Explain linked Allocation of file in detail.

c) Explain iphone Architecture in detail.

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Q5) Attempt any one of the following: **[1×3=3]**

a) Write Advantages and Disadvantage of Distributed systems.

b) Assume there are total 200 tracks are presents on each surface of the disk. If request quecee is 30,140,20,170,60,190 and initial position of the head is 120. Apply following disk scheduling Algorithms & calculate total head movement.

i) FCFS

ii) SSTF

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Total No. of Questions : 5]

SEAT No. :

PD509

[Total No. of Pages : 3

[6468]-61
T.Y.B.Sc. (Computer Science)
CS- 361 :OPERATING SYSTEMS - II
(Revised 2019 Pattern) (Semester - VI)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***Assume suitable data if necessary.***
3) ***Figures to the right indicate full marks.***

Q1) Attempt any EIGHT of the following: **[8×1=8]**

a) What is the name of the direct edge from resource Ri to process Pi in resource allocation graph?

b) Which necessary condition of deadlock does Resource Ordering forprevention of deadlock?

c) What is absolute path name in directory structure?

d) Which file operation involves removing all data from a file, leaving the file empty?

e) Which disk scheduling algorithm minimized the average seek time over a number of disk arm movements.

f) What is size scalability in distributed systems?

g) List any two types of cloud.

h) What is distributed transaction processing?

i) Write any two special constrain of mobile operating system.

j) Define native level programming.

Q2) Attempt any FOUR of the following. **[4×2=8]**

a) If process P1 is holding resource R1 and waiting for resource R2 held by process P2, while process P2 is holding resource R2 and waiting for resource R1 held by process P1. Draw the resource allocation graph and check that system is in deadlock state or not.

b) Explain any two disk performance parameters.

P.T.O.

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c) What is directory? List any two types of directory.

d) How does grid computing differ from cluster computing?

e) Explain in brief, any two special service requirements of mobile oprating system.

Q3) Attempt any TWO of the following: **[2×4=8]**

a) Consider a system with 5 processes {P1, P2, P3, P4, P5} and three resource types {A, B, C}. The allocation and maximum demand matrices are as follows:

	Allocation		
	A	B	C
P1	2	1	0
P2	2	0	1
P3	2	1	2
P4	2	1	1
P5	0	0	2

	Max		
	A	B	C
P1	6	5	3
P2	3	2	2
P3	7	1	2
P4	2	2	2
P5	4	3	3

Available		
A	B	C
2	2	1

Answer the following questions using Banker's Algorithm

i) What are the contains of Need matrix?

ii) Is the system in safe state? If yes, give the safe sequence.

b) Explain contiguous and linked file allocation methods.

c) Differentiate between ARM architecture and x86 architecture.

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Q4) Attempt any TWO of the following: **[2×4=8]**

a) Explain any four free space management methods used in file system management.

b) Explain deadlock recovery methods in details.

c) List down the architectural styles in distributed operating system & explain any one in detail.

Q5) Attempt any ONE of the following: **[1×3=3]**

a) Suppose that a disk drive has 200 cylinders, numberd 0 to 199 and order of request is: (82, 170, 43, 140, 24, 16, 190). And current position of Read/Write head is : 50. Starting from the current head position, what is the total distance that the disk arm moves to satisfy all the pending requests, for FCFS/FIFO disk scheduling Algorithms?

b) Explain the benefits or advantages of distributed systems.

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